

PayReality

Enabling enterprises to safely delegate business authority
to AI systems through policy controls and verifiable decision records.

SAFE Note · \$150K · OPA/Rego · Rust Engine · Dual Ed25519 VIC · MVP Built · Patent Pending PPN00002476



The Execution Vacuum

THE INFRASTRUCTURE HAS MOVED

Visa, Stripe, Google, and AWS have already shipped agentic commerce APIs. Autonomous procurement agents can now legally connect to live payment rails, issue invoices, execute settlements, and route treasury — today, without human sign-off on each transaction.

WHAT AGENTS CAN NOW DO

- Autonomous B2B procurement — multi-vendor, multi-currency
- Supply chain sourcing — real-time bid evaluation and commitment
- Treasury routing — cross-border settlement execution
- Automated accounts payable — invoice matching and disbursement

ENTERPRISE RISK COMMITTEES ARE BLOCKING DEPLOYMENT

AI models are **probabilistic**. They hallucinate. They misinterpret instructions. They can be prompt-injected. When a probabilistic model controls a deterministic payment rail, a single mutated API payload produces an **immediate, un-reversible balance sheet liability**.

THE UNANSWERABLE QUESTION

"If the AI agent hallucinated a beneficiary and sent \$500,000 to the wrong account — can you prove, cryptographically, what policy it was operating under at the exact moment of execution?"

- No independent audit trail of AI intent exists pre-settlement
- Application logs prove a transaction happened — not *why*
- Insurance claims cannot be settled without machine-verifiable evidence

⚡ **The gap is not the AI capability.** Infrastructure giants have solved execution. The gap is the **accountability layer** — a verifiable, tamper-proof record of what the AI was authorised to do and what it actually did.

The Regulatory Catalyst

Aug 2, 2026

High-Risk Autonomous Systems

ART. 12

Continuous Logging & Traceability

High-risk AI systems must automatically generate logs comprehensive enough for post-hoc regulatory review. Logs must capture **every decision**, every input, and the state of the system at the moment of execution.

VIC MAPPING

PayReality's VIC captures agent identity, policy envelope ID, rule evaluation results, raw payload, enforcement action, and human reviewer decision — all cryptographically timestamped before settlement.

ART. 14

Human Oversight & Override

Enterprises must be able to **intervene, override, or halt** autonomous AI decisions in real time. Oversight must be verifiable — not just claimed. A post-hoc audit trail of human approval or block decisions is legally required.

VIC MAPPING

Every PayReality **HOLD_FOR_HUMAN_REVIEW** state logs the triggering rule, the escalation event, and the **HUMAN_REMEDIATION_DECISION** taken — sealed into the VIC before the transaction resolves.

LEGAL POSITION

Independent cryptographic verification is no longer a *technical preference* — it is a **mandatory legal requirement** for enterprise banking and corporate runtimes operating AI systems after August 2, 2026.

SCOPE

- EU enterprises operating autonomous financial agents
- Non-EU entities processing EU counterparty transactions
- Insurers underwriting AI performance liability products

The Trust & Evidence Layer

Probabilistic AI Layer

o PayReality - Platform-Neutral Cryptographic Notary

Corporate Transaction Rails

01 - POLICY CONTROLS

Deterministic Authority Delegation

Human compliance officers define **Rego policy rules** — compiled from SOPs, board mandates, and risk frameworks. These rules are the *only* execution path. The AI agent's output is evaluated against them before any action reaches a rail. No LLM on the enforcement path.

02 - RUNTIME ENFORCEMENT

Sub-Millisecond Inline Intercept

A **Rust inline interceptor** executes OPA/Rego evaluation in-process at sub-millisecond speeds. Every payload is validated before touching a gateway. Zero silent fallbacks — ambiguous intent triggers an immediate **HOLD_FOR_HUMAN_REVIEW** state.

03 - VERIFIABLE EVIDENCE

Cryptographic VIC Receipts

Every enforcement decision mints a **Verifiable Intent Certificate** — dual-signed (Platform + Agent) Ed25519 signatures, pushed to **AWS S3 Object Lock** (WORM storage). Independently verifiable by anyone with the public key.

WHY PLATFORM-NEUTRAL MATTERS

Courts, compliance officers, and insurance syndicates require a **neutral, third-party notary** whose signatures can be verified independently — not an audit generated by the same hyperscaler running the AI system. PayReality sits outside all cloud provider trust boundaries.

TECHNICAL PROPERTIES

- No runtime LLMs on the enforcement execution path
- Dual Ed25519 signatures — Platform key + Agent key
- AWS S3 Object Lock (WORM) — immutable storage
- Independently verifiable — public key verification
- No silent fallbacks — ambiguity always escalates to human

The 3-Step Technical Pipeline

01 POLICY COMPILER

Human Intent → Rego Code

Compliance PDFs, board mandates, and procurement SOPs are translated into executable **OPA/Rego policy rules** by compliance officers using the Policy Manager UI. Rules are versioned, auditable, and human-readable.

```
# Compiled Rego rule - max disbursement
deny {
  input.payload.amount_cents > 10000000
  not input.approvals.cfo_signed
}
```

02 RUNTIME ENFORCEMENT

Rust + OPA In-Process

Every AI-initiated payload is intercepted by the **Rust inline engine** before any gateway contact. OPA/Rego evaluation runs in-process — no network round-trips, sub-millisecond latency. Three outcomes only: **APPROVED**, **BLOCKED**, or **HOLD**.

Vendor whitelisted : amount within limit	APPROVED
Amount exceeds R100,000 threshold	BLOCKED
Intent unresolvable — missing provenance	HOLD → HUMAN

03 VERIFIABLE INTENT CERTIFICATE

Immutable Evidence Receipt

Every decision — approve, block, or hold — mints a **dual-signed VIC**. Platform Ed25519 key + Agent Ed25519 key. Pushed to **S3 Object Lock** (WORM). Verifiable by any party holding the public key — courts, auditors, insurers, regulators.

```
"enforcement_action": "BLOCKED",
"policy_matched": "max_disbursement_v3",
"dual_sig": {
  "platform": "Ed25519:a3f0...",
  "agent": "Ed25519:7c2b..."
},
"storage": "s3://vic-registry/WORM"
```

High-Stakes Transactions First

Our beachhead is locked onto the transaction categories where a single compromised AI payload produces an **immediate, quantifiable financial crisis** — giving enterprise risk committees the clearest possible ROI case for deployment.

WHY NOT BROADER FIRST

Low-stakes AI actions (drafting emails, scheduling) have low accountability pressure. **High-stakes financial execution** has regulators, auditors, boards, and insurers all demanding the same answer simultaneously. The urgency justifies immediate deployment.

DISTRIBUTION MODEL

Insurance underwriters **mandate PayReality** as a condition of activating AI performance liability policies — pulling us into enterprise deployments without a direct enterprise sales cycle.

IMMEDIATE FOCUS SEGMENTS



Automated B2B Procurement

Autonomous agents issuing purchase orders to multi-vendor supply chains. A corrupted payload routes capital to a fraudster's account — undetectable until the legitimate supplier escalates.

Winter Pilot Target



Supply Chain Sourcing

Real-time bid evaluation and commitment by AI agents across cross-border suppliers. Contract execution on hallucinated terms creates immediate legal and financial liability across multiple jurisdictions.

Q1 2027 Target



Financial Settlements & Treasury Routing

Cross-border treasury automation — highest single-transaction value and the most severe consequence of an unverified AI decision. Regulators are watching this category first.

Regulatory Priority

Insurable AI

THE UNDERWRITING PROBLEM

Insurance carriers launching AI performance liability products face a structural obstacle: **they cannot audit the internal code of every enterprise AI deployment** they underwrite. The codebase complexity is unbounded, proprietary, and constantly changing.

WHAT INSURERS ACTUALLY NEED

An **independent, un-falsifiable telemetry standard** — a neutral VIC registry whose cryptographic signatures can be verified without trusting any party's internal infrastructure. The insurer verifies the seal, not the code.

- ✔ Verify the Platform Ed25519 signature against PayReality's public key
- ✔ Verify the Agent Ed25519 signature against the enterprise's registered agent key
- ✔ Confirm WORM hash — the VIC has not been altered since creation
- ✔ Policy envelope ID maps to the version the enterprise declared to the insurer

THE DISTRIBUTION FLYWHEEL



AISURE DISCOVERY STATUS

Active technical discovery with AISure initiative to evaluate PayReality as the core telemetry standard for their automated transaction liability product line. Status: **pre-contract discovery and architecture review**.

STRATEGIC IMPLICATION

A single insurer mandate creates a pull channel to *every enterprise in their book of business* — without PayReality running a direct enterprise sales motion against each account individually.

Why Big Tech Cannot Own This

Microsoft Azure AI Studio, AWS Bedrock, and OpenAI can audit their own environments. External courts, compliance officers, and insurance syndicates require something different: **a neutral third-party notary whose trust boundary is structurally separate from the cloud provider running the AI.**

THE HYPERSCALER CONFLICT

A hyperscaler-generated audit log certifies the hyperscaler's own infrastructure. When a dispute arises between an enterprise and a counterparty, no independent judge or insurer accepts a self-generated audit as neutral evidence. **You cannot be both the cloud and the notary.**

STRUCTURAL MOAT COMPONENTS



Platform Neutrality

PayReality runs alongside any cloud AI stack — AWS, Azure, GCP, on-premise. No vendor lock-in for the enterprise. The notary function is structurally independent of the execution environment being audited.



Legal Admissibility Standard

Dual Ed25519 signatures + S3 Object Lock WORM create a chain of custody designed to meet evidentiary standards in commercial dispute proceedings — not just IT audit requirements.



No Runtime LLM on the Enforcement Path

Deterministic Rego execution means the enforcement engine itself cannot hallucinate or drift. There is no probabilistic component that a hyperscaler can replicate with a larger model — the moat is architectural.



Provisional Patent + First-Mover IP

PPN00002476 protects the dual-signed VIC schema and runtime policy adaptation framework. PCT international application planned Q2 2027. Insurer mandates create switching costs before a competitor can replicate the architecture.

Execution DNA

CEO & FOUNDER

Sean Chihwendu

Procurement & Frontend Arch.

\$2.5M USD B2G procurement management. Identified the enterprise "shadow workflow" gap first-hand. Architected PayReality's Next.js + Tailwind policy management environment and investor narrative.

Equity **40.5%**

CO-FOUNDER & CTO

Chukwudi Nathan Obiekwe

Systems & Cryptography

Ex-RisCura, Turing, Linum Labs. Built PayReality's FastAPI backend, real Ed25519 cryptographic layer, and the OPA/Rego policy evaluation runtime. Leads all core Rust engine development.

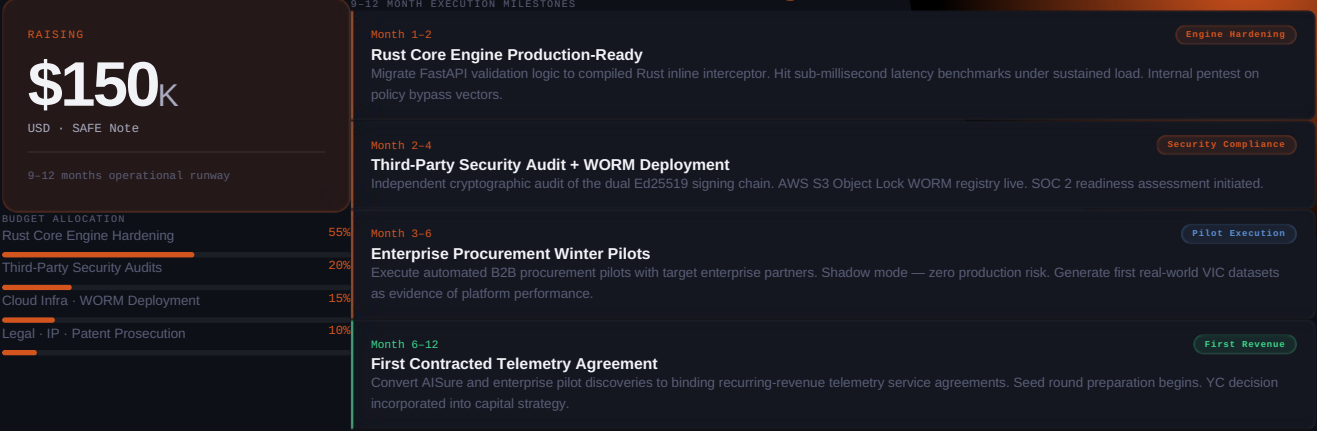
Equity **27.0%**

100% full-time. Both co-founders have no secondary obligations. PayReality is our  single professional focus. 22.5% equity pool unallocated — available for incoming senior engineer or advisor.

INFRASTRUCTURE VALIDATION — JUNE 2026

FastAPI Python backend — local validation	BUILT
OPA/Rego policy evaluation runtime	BUILT
Dual Ed25519 VIC cryptographic signing	BUILT
Next.js + Tailwind Policy Manager UI	BUILT
Provisional Patent PPN00002476	FILED
Rust core engine hardening	POST-FUNDING
AWS S3 Object Lock WORM deployment	POST-FUNDING
YC application — decision pending	SUBMITTED

\$150K SAFE · 9–12 Month Runway



Month 1–2
Rust Core Engine Production-Ready
Migrate FastAPI validation logic to compiled Rust inline interceptor. Hit sub-millisecond latency benchmarks under sustained load. Internal pentest on policy bypass vectors.

Month 2–4
Third-Party Security Audit + WORM Deployment
Independent cryptographic audit of the dual Ed25519 signing chain. AWS S3 Object Lock WORM registry live. SOC 2 readiness assessment initiated.

Month 3–6
Enterprise Procurement Winter Pilots
Execute automated B2B procurement pilots with target enterprise partners. Shadow mode — zero production risk. Generate first real-world VIC datasets as evidence of platform performance.

Month 6–12
First Contracted Telemetry Agreement
Convert AISure and enterprise pilot discoveries to binding recurring-revenue telemetry service agreements. Seed round preparation begins. YC decision incorporated into capital strategy.